Shaohu Zhang Ph.D.

Assistant Professor, Department of Computer Science

North Carolina A&T State University

szhang1@ncat.edu | zhangshaohu.github.io

RESEARCH INTERESTS

- **Trustworthy & Explainable AI:** designing context-aware mobile systems to bridge CPS and AI for reliable and adaptive interactions.
- **Cybersecurity:** fingerprinting and human biometric security, with a focus on safeguarding privacy and authentication through embedded sensors in digital devices.
- **Mobile Computing/Sensing Systems:** sensing with ubiquitous modalities, such as WiFi, camera, IMU, and sound in intelligent transportation and smart homes.
- **LLM and NLP:** speech privacy, privacy policy and compliance, and LLM applications in built environment and agriculture.

APPOINTMENTS & PROFESSIONAL EXPERIENCE

Assistant Professor Aug 2025–present
Department of Computer Science, North Carolina A&T State University

Assistant Professor Aug 2023–Jun 2025
Department of Math & Computer Science, University of North Carolina at Pembroke

Graduate Research / Teaching Assistant Jan 2020–May 2023
Wolfpack Security and Privacy Research (WSPR) Lab, North Carolina State University, Raleigh, NC
Digital security and privacy with Prof. Anupam Das.

Graduate Research / Teaching Assistant Aug 2017–Dec 2019
Wolfpack Interactive, Sensing and Networking (WiSN) Lab, North Carolina State University, Raleigh, NC
WiFi sensing with Prof. Muhammad Shahzad.

Graduate Research Assistant Aug 2015–Jul 2017
Department of Civil and Environmental Engineering, South Dakota State University, Brookings, SD
Transportation safety with Prof. Xiao Qin and Prof. Jonathan Wood.

Graduate Research Assistant Aug 2015–Dec 2016
Department of Computer Science and Electrical Engineering, South Dakota State University, Brookings, SD
Mobile systems with Prof. Wei Wang and Prof. Myounggyu Won.

Logistics Engineer Jun 2012–Aug 2013
China Railway Materials Commercial Corp, Shanghai, China

Marketing Research Intern Apr 2012–May 2012
SITC Logistics, Shanghai, China

Instructor Sep 2011–Jul 2012
Shanghai Maritime University, Shanghai, China

EDUCATION

Ph.D., Computer Science 2023
North Carolina State University, Raleigh, NC

M.S., Civil Engineering 2017
South Dakota State University, Brookings, SD
Computer Science, M.S. non-degree study.

B.A., Marine Fishery Science and Technology 2010
Zhejiang Ocean University, Zhoushan, China

AWARDED RESEARCH FUNDING

AI-WeedSight: Advancing Precision and Sustainable Weed Management with Robotics and Drone Imaging
Co-PI; PI: Dr. Harmandeep Sharma
North Carolina Collaboratory | Total award: $105,514
Jan–Dec 2026

Multidisciplinary Smart Weather Toolkit for Small-scale and BIPOC Farmers
Co-PI; PI: Dr. Crystal Marshall
USDA NIFA Evans-Allen Grant | Total award: $650,000
Oct 2023–Sep 2026 | Fund #: NC.X 374-5-24-170-1

Cyber Escape Rooms: Fostering Student Engagement and Boosting Student Prospects Enrollment in Cybersecurity and Computer Science Programs
Co-PI; PI: Dr. Prashanth Busireddygari
UNCP | Total award: $10,000
Jan 1–May 31, 2024

Neighborhood Microscope: Leveraging Big Data Sources to Examine Health and Well-being in Urban and Rural NC Neighborhoods
PI; Co-PI: Dr. Melanie Escue
North Carolina Collaboratory | Total award: $86,000
Jan 1, 2024–Jun 30, 2025 | Fund #: Collab_486

PUBLICATIONS

J Journal; **C** Conference; **W** Workshop. Underlined authors are advisees.

[17] **(J)** Subhram Dasgupta, Pal Dave, Md Tasnim Alam, Kushal Badal, **Shaohu Zhang**, Hamidreza Moradi, Madhuri Siddula, and Kaushik Roy. "Vision-Language Models in Medical Applications: A Review." IEEE Access, vol. 14, pp. 92070–92099, 2026. DOI: 10.1109/ACCESS.2026.3704901.

[16] **(W)** Najmul Hasan, Prashanth BusiReddyGari, Haitao Zhao, Yihao Ren, Jinsheng Xu, **Shaohu Zhang**. LLMPEA: Leveraging Large Language Models Against Phishing Email Attacks. 2026 56th Annual IEEE/IFIP International Conference on Dependable Systems and Networks Workshop on Dependable and Secure Machine Learning (DSN-DSML'26).

[15] **(C)** Yihao Ren, Pan Lu, Lu Gao, **Shaohu Zhang**. A Comparative Study of Multimodel Large Language Models for Tire Size Recognition from Real-World Images, Transportation Research Board 2026 Annual Meeting. Washington DC.

[14] **(J)** Ruoyu Zhao, Xiantao Jiang, F. Richard Yu, Victor C.M. Leung, Tao Wang, and **Shaohu Zhang**. Leveraging Cross-Attention Transformer and Multi-Feature Fusion for Cross-Linguistic Speech Emotion Recognition. IEEE Internet of Things Journal, 2025 (CiteScore: Q1). [Paper]

[13] **(W)** Andrew Cart, **Shaohu Zhang**, Melanie Escue, Xugui Zhou, Haitao Zhao, Prashanth Busireddygari, Beiyu Lin, Shuang Li. Decoding Neighborhood Environments with Large Language Models. 2025 55th Annual IEEE/IFIP International Conference on Dependable Systems and Networks Workshop on Dependable and Secure Machine Learning (DSN-DSML'25)

[12] **(C)** Zhouyu Li, Ruozhou Yu, Anupam Das, **Shaohu Zhang**, Huayue Gu, Xiaojian Wang, Fangtong Zhou, Aafaq Sabir, Dilawer Ahmed, and Ahsan Zafar. INSPIRE: Instance-level Privacy-preserving Transformation for Vehicular Camera Videos. Proceedings of the 32nd International Conference on Computer Communications and Networks. **IEEE ICCCN'23** (CORE ranking: B, acceptance rate: 55/181=30%).

[11] **(C) Shaohu Zhang**, Zhouyu Li, Anupam Das. VoicePM: A Robust Privacy Measurement on Voice Anonymity, *16th ACM Conference on Security and Privacy in Wireless and Mobile Networks.* **ACM WiSec 2023** (CORE ranking: B, acceptance rate: 34/134 =25%).

[10] **(C) Shaohu Zhang**, Aafaq Sabir, Anupam Das. Speaker Orientation-Aware Privacy Control to Thwart Misactivation of Voice Assistants, *The 53rd Annual IEEE/IFIP International Conference on Dependable Systems and Network.* **IEEE DSN 2023** (CORE ranking: A, acceptance rate: 47/235=20%).

[9] **(C) Shaohu Zhang**, Anupam Das. Enabling 2-FA for Smart Home Voice Assistants using Inaudible Acoustic Signal, *In 24th International Symposium on Research in Attacks, Intrusions and Defenses, pp. 251-265. 2021.* **RAID'21** (CORE ranking: A, acceptance rate: 33/138=23.9%).

[8] **(J)** Jonathan Wood, **Shaohu Zhang**. Evaluating Relationships Between Perception-Reaction Times, Emergency Deceleration Rates, and Crash Outcomes using Naturalistic Driving Data. *Journal of Transportation research record 2675, no. 1 (2021): 213-223.* **TRR'21** (CiteScore: Q2, acceptance rate: 20%).

[7] **(C) Shaohu Zhang**, Raghav Venkatnarayan, Muhammad Shahzad. A WiFi-based Home Security System. *In 2020 IEEE 17th International Conference on Mobile Ad Hoc and Sensor Systems (MASS), pp. 129-137.* **IEEE MASS'20** (CORE ranking: B, acceptance rate:28%).

[6] **(J)** Jonathan Wood, **Shaohu Zhang**. Identification and Calculation of Horizontal Curves for Low-Volume Roadways using Smartphone Sensors. *Journal of Transportation Research Record, 2672(39), 1-10. 2018.* **TRR'18** (CiteScore: Q2, acceptance rate: 20%).

[5] **(C+J)** Muhammad Shahzad, **Shaohu Zhang**. Augmenting User Identification with WiFi Based Gesture Recognition. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, 2(3), pp.1-27, 2018.* **IMWUT/Ubicomp'18** (CORE ranking: A*, acceptance rate:18%).

[4] **(C) Shaohu Zhang**, Myounggyu Won, Sang H. Son. Low-cost and Non-intrusive Traffic Monitoring System Using WiFi. *In 2017 26th International Conference on Computer Communication and Networks (ICCCN), pp. 1-9. IEEE, 2017.* **ICCCN'17** (CORE ranking: B, acceptance rate:28.6%).

[3] **(C)** Myounggyu Won, **Shaohu Zhang**, Appala Chekuri, Sang H. Son. Enabling Energy-Efficient Driving Route Detection Using a Built-in Smartphone Barometer Sensor, *In 2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC), pp. 2378-2385. IEEE, 2016.*

[2] **(C) Shaohu Zhang**, Myounggyu Won, Sang H. Son. Low-cost Realtime Horizontal Curve Detection Using Inertial Sensors of a Smartphone. *In 2016 IEEE 84th Vehicular Technology Conference (VTC-Fall), pp. 1-5. IEEE, 2016.* (CORE ranking: B).

[1] **(C)** Xiao Qin, **Shaohu Zhang**, Wei Wang. Advanced Curve-speed Warning System Using an In-Vehicle Head-Up Display. *Proceedings of 94th Transportation Research Board Meeting, Washington, D.C, 2015.*

MANUSCRIPTS UNDER REVIEW

[5] Yihao Ren, **Shaohu Zhang** et. al. SR-OCR: A Spatial Robustness-Oriented Evaluation Framework for Training-Free Tire Size Recognition Using Multimodal Large Language Models. Journal of Transportation Research Record. (Under Review)

[4] Haitao Zhao, **Shaohu Zhang** et al. A Position- and Syntax- Aware Transformer Framework for Mathematical Expression Recognition. IEEE Transactions on Artificial Intelligence. (Under Review)

[3] Md Tasnim Alam, Subhram Dasgupta, Kushal Badal, Raihana Tasnim, Xiaohong Yuan, Kaushik Roy, Madhuri Siddula, and **Shaohu Zhang**, From Black Box to Trusted Guardian: A Comprehensive Survey on Explainable Artificial Intelligence for Human Action Recognition in Safety-Critical Applications. IEEE Access (Under Review).

[2] Mesafint Fanuel, Mahmoud Nabil Mahmoud, Crystal Cook Marshal, Vishal Lakhotia, Biswanath Dari, Kaushik Roy, **Shaohu Zhang**, Region-Aware Knowledge-Based RAG System for Agriculture, IEEE Access (Under Review).

[1] **(J) Shaohu Zhang**, Andrew Cart, Melaine Escue, Yuhang Jiang, Emmanuel Obeng-Gyasi, Meng Zhao, Kai Zhang, Shuang Li, Seeing the Streets with GenAI: Decoding Built Environment Using LLMs, ACM Transactions on Intelligent Systems and Technology (Under Review).

POSTERS & SHORT PAPERS

[5] Bontu Geremew, Sharun Akter Khushbu, **Shaohu Zhang**. Edge-Based Privacy Auditing for Trustworthy Humanoid Robot Interaction. 2026 56th Annual IEEE/IFIP International Conference on Dependable Systems and Networks, Charlotte, USA, June 22-25, 2026.

[4] **Shaohu Zhang**, Zhouyu Li, Anupam Das. Privacy Measurement of Physical Attributes on Voice Anonymity, *The 30th Annual International Conference on Mobile Computing and Networking Nov. 18-22, 2024, Washington, D.C., USA.* **MobiCom'24**.

[3] **Shaohu Zhang**, Aafaq Sabir, Anupam Das. POSTER: Enhancing Security and Privacy Control for Voice Assistants Using Speaker Orientation, *16th ACM Conference on Security and Privacy in Wireless and Mobile Networks.* **WiSec'23**.

[2] **Shaohu Zhang**, Anupam Das. A 2-FA for home voice assistants using inaudible acoustic signal. *In Proceedings of the 27th Annual International Conference on Mobile Computing and Networking, pp. 834-836. 2021.* **MobiCom'21**.

[1] **Shaohu Zhang**, Myounggyu Won, Sang H. Son. WiTraffic: Non-intrusive Vehicle Classification Using WiFi. *In Proceedings of the 14th ACM Conference on Embedded Network Sensor Systems, pp. 358-359. 2016.* **SenSys'16**

TECHNICAL REPORT

[1] Jonathan Wood *, **Shaohu Zhang**. Evaluating Relationships Between Perception-Reaction Times, Emergency Deceleration Rates, and Crash Outcomes Using Naturalistic Driving Data. *MPC-17-338, North Dakota State University - Upper Great Plains Transportation Institute, Fargo: Mountain-Plains Consortium, 2017.*

TEACHING EXPERIENCE

North Carolina A&T State University, Greensboro, NC

COMP 895 LLM and Security — Fall 2026

COMP 710 Software Specifications and Design — Spring 2026

COMP 385 Theory of Computing — Fall 2025, Spring 2026

University of North Carolina at Pembroke, Pembroke, NC

CSC 2920 Software Engineering (in-person & online) — Spring 2024, Fall 2024, Spring 2025
CYB 4030 Introduction to Digital Forensics — Spring 2024, Spring 2025
CYB 4020 Network Security — Fall 2024
CSC 1750 Introduction to Algorithms — Fall 2023
CSC 1760 Introduction to Programming — Fall 2023
CSC 1850 Object-oriented Programming — Fall 2023

Teaching Assistant / Lab Instructor

North Carolina State University, Raleigh, NC

CSC/ECE 773 Advanced Internet Protocol — Spring 2023
Instructor: Prof. Khaled Harfoush

CSC/ECE 573/591 Internet Protocol — Fall 2022
Instructor: Prof. Khaled Harfoush

CSC 433 Privacy in the Digital Age — Spring 2021
Instructor: Prof. Anupam Das

CSC 533 Privacy in the Digital Age — Fall 2020
Instructor: Prof. Anupam Das

CSC/ECE 591/791 Internet of Things — Spring 2020
Instructor: Prof. Muhammad Shahzad

CSC/ECE 573/591 Internet Protocol — Spring 2020
Instructor: Prof. Muhammad Shahzad

CSC 453 Internet of Things — Spring 2019
Instructor: Prof. Muhammad Shahzad

STUDENTS & MENTORING

Research Associate

Dr. Mesafint Fanuel, 2025–present. Topic: Agriculture AI agent.

Current Ph.D. Students

Mr. Md Mehraz Hosen, Fall 2026–present. Topic: Smart Transportation.

Ms. Bontu Geremew, Spring 2026–present. Topic: Humanoid Robot.

Ms. Sharun Akter Khushbu, Spring 2026–present. Topic: Agriculture AI.

Current Ph.D. Committees

Kushal Badal, CS. Advised by Dr. Xiaohong Yuan.

Anurodh Jha, Applied Science and Technology. Advised by Dr. Emmanuel Obeng-Gyasi.

Shaghayegh Shajarian, CS. Advised by Dr. Sajad Khorsandroo and Dr. Mahmoud Abdelsalam.

Past Ph.D. Committees

Pal Dave, "Multimodal Cyberbullying Detection: From Deep Learning to Agentic AI." CS; advised by Dr. Xiaohong Yuan. Graduated Summer 2026.

Dagmawit Tadesse Aga, "Advancing Privacy-Preserving Multimodal Federated Learning for IoMT: From Foundational Research to Deployment-Ready Clinical Technology." CS; advised by Dr. Madhuri Siddula. Graduated Summer 2026.

Current Master's Students

Mr. Ruoyu Zhao, CS, SMU. Topic: Speech emotion.

Current Undergraduate Students

Mr. Charlestone Mayenga, B.S. Computer Science. Topic: Privacy Measurement on Motion Anonymity.

Mr. Adan Maldonado-Hernandez, B.S. Computer Engineering. Topic: Farm sensor system, AI in Agriculture.

Mr. Bryson Wingate, B.S. Computer Science. Topic: Robotics and Drone Imaging.

Ms. Coraline Song, B.S. Computer Science, UNCC. Topic: LLM privacy.

Past Undergraduate Students

Mr. Mohammad Kawsar, B.S. Computer Science, UNCP. Mobicom 2025 Travel Grant; Dr. Charles Humphrey Undergraduate Conference Travel Award.

Mr. Najmul Hasan, B.S. Computer Science, UNCP. Spring 2024 Semester Long Undergraduate Research Fellowship; UNCP Summer 2025 Undergraduate Research Fellowship.

Mr. Andrew Cart, B.S. Computer Science, UNCP. Graduated in 2025; Microsoft Software Engineer.

High School Students

Amogh Prashanth, NCAT Early College, 2026.

Ricky Zhang, Greensboro Day School, 2026.

Ernest Hu, Early College at Guilford, 2026.

Alan Cai, North Carolina School of Science and Mathematics – Durham, 2026.

HONORS & AWARDS

- **Mentored Teaching Fellowship**, College of Engineering, North Carolina State University, 2023.
- **2022 Summer Graduate Fellowship**, College of Engineering, North Carolina State University, 2022.
- **Enhancement Fee Travel Award**, College of Engineering, North Carolina State University, 2020, 2023.
- **Graduate Research Award**, College of Engineering, North Carolina State University, 2018.
- **Student Travel Grant**, HotMobile'23, RAID'21, CCS'21, MobiCom'21, MobiCom'17, and SenSys'16.
- **Sigma Xi Graduate Research Award**, South Dakota State University, 2016.
- **Undergraduate Thesis Award**, Zhejiang Ocean University, China, 2010.

INVITED TALKS & PRESENTATIONS

[18] AgriRegion: Region-Aware Retrieval for High-Fidelity Agricultural Advice, 2026 AI in Agriculture Conference, NCSU, March 31, 2026

[17] Seeing the Streets with GenAI: Decoding Built Environment Using LLMs, Morgan State University, March 27, 2026

[16] Insights from an Interdisciplinary NC Collaboratory Project: The Influence of the Built Environment on Health in NC, 11th Annual Demography Daze, Duke University Population Research Institute (DUPRI) and Carolina Population Center (CPC), September 26, 2025

[15] Seeing the Streets with GenAI: Decoding Built Environment Using LLMs, The 2025 Artificial Intelligence Conference (NCAT), Greensboro NC,September 25, 2025

[14] Human-Centered Cybersecurity: Designing Context-Aware and Trustworthy Solutions for Smart Ecosystems, North Carolina Agricultural and Technical State University, March 2025.

[13] Human-Centered Cybersecurity: Designing Context-Aware and Trustworthy Solutions for Smart Ecosystems, Louisiana State University, March 2025.

[12] Towards Context-aware and Trustworthy Voice Assistants, Thomas College of Business and Economics at UNCP, October 2024.

[11] Speaker Orientation-Aware Privacy Control to Thwart Misactivation of Voice Assistants, The 53rd Annual IEEE/IFIP International Conference on Dependable Systems and Network, IEEE DSN 2023.

[10] VoicePM: A Robust Privacy Measurement on Voice Anonymity, *16th ACM Conference on Security and Privacy in Wireless and Mobile Networks.* WiSec 2023 virtual.

[9] Web Privacy in the Digital World. University of North Carolina, Pembroke, March 2023.

[8] Enabling 2-FA for Smart Home Voice Assistants using Inaudible Acoustic Signal. *The 24th International Symposium on Research in Attacks, Intrusions and Defenses.* (RAID'21 virtual).

[7] 2-FA for Smart Home Voice Assistants using Inaudible Acoustic Signal. *In Proceedings of the 27th Annual International Conference on Mobile Computing and Networking.* (Mobicom'21 virtual).

[6] Security on Android Devices. *NCSU Data Privacy Month 2021.*

[5] A WiFi-based Home Security System. *The 17th IEEE International Conference on Mobile Ad Hoc and Sensor Systems.* (MASS'20 virtual).

[4] Identification and Calculation of Horizontal Curves for Low-Volume Roadways using Smartphone Sensors. *In the 97th Transportation Research Board Annual Meeting (TRB'18), Washington D.C.*

[3] WiTraffic: Non-intrusive Vehicle Classification Using WiFi. *In the 14th ACM Conference on Embedded Networked Sensor Systems (SenSys'16), Stanford University.*

[2] Horizontal Curve Detection Using Inertial Sensors of a Smartphone. *Sigma Xi Chapter, South Dakota State University, 2016*

[1] Avoiding Roadway Departure Crashes with an In-Vehicle Head-Up Display. *In the TRB 94th Transportation Research Board Annual Meeting (TRB'15) Washington D.C.*

PROFESSIONAL ACTIVITIES

Professional memberships: IEEE, ACM.

Panel

- NSF Panelist, 2026.

Co-Chair

- Co-chair, Privacy Check-up Sessions, NCSU Data Privacy Month 2021. North Carolina State University, February 2021.

Program Committee

- IEEE Global Communications Conference (GLOBECOM), 2026.
- The 26th Privacy Enhancing Technologies Symposium (PETS 2026).
- 8th Dependable and Secure Machine Learning Workshop (DSML 2025, co-hosted with DSN 2025), Naples, Italy, June 23–26, 2025.
- 2025 International Conference on Computing, Networking and Communications (ICNC 2025), Honolulu, Hawaii.
- The 25th Privacy Enhancing Technologies Symposium (PETS 2025), Washington, DC.
- The First International Workshop on IoT Datasets for Multi-modal Large Model (IoT-MMLM 2024), in conjunction with SenSys 2024, November 4–7, 2024, Hangzhou, China.

- The 2024 IEEE 100th Vehicular Technology Conference (VTC2024-Fall), Washington, DC.
- The 2nd ACM Workshop on Smart Wearable Systems and Applications (SmartWear2023), in conjunction with MobiCom 2023, October 6, 2023, Madrid, Spain.
- The 2023 IEEE 98th Vehicular Technology Conference (VTC2023-Fall), Hong Kong.
- Ph.D. Forum, The 53rd Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN'23).

Artifact Review Program Committee

- USENIX Security Symposium, 2023.
- Privacy Enhancing Technologies Symposium (PoPETs), 2023.
- ACM ASIA Conference on Computer and Communications Security (AsiaCCS), 2023.
- Annual Computer Security Applications Conference (ACSAC), 2023.
- ACM Conference on Security and Privacy in Wireless and Mobile Networks (ACM WiSec), 2022–2023.

Community Service

- Judge, Region 4 North Carolina Science and Engineering Fair, 2024, 2025.
- Judge, North Carolina Science and Engineering Fair (NCSEF), 2024, 2025.

University Service

- Cabin Road Endowed Professor of Artificial Intelligence hiring committee, 2026.
- Advisory Board of the Cyber Security Program at Fayetteville State University, 2026–present.
- Cybersecurity faculty hiring committee, UNCP, 2025.

Conference Reviewer

- Privacy Enhancing Technologies Symposium (PETS), 2023–2026.
- ACM Conference on Computer and Communications Security (CCS), 2021–2023.
- Transportation Research Board Annual Meeting (TRB), 2023–2026.
- Symposium on Security and Privacy (IEEE S&P), 2021, 2024.
- ISOC Networked and Distributed System Security Symposium (NDSS), 2021–2023.
- Annual Computer Security Applications Conference (ACSAC), 2022–2023.
- Security and Privacy in Wireless and Mobile Networks (WiSec), 2021–2024.
- ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys), 2019.
- Proceedings of Transportation Research Board Meeting, 2023–2026.

Journal Reviewer

- International Journal of Human-Computer Interaction (IJHCI).
- ACM Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT).
- IEEE Internet of Things Journal.
- ACM Transactions on Interactive Intelligent Systems.
- IEEE Transactions on Mobile Computing.
- IEEE Transactions on Reliability.
- IEEE Security and Privacy.